

ISO 25010:

RELIABILITY & FLEXIBILITY

Group 5



RELIABILITY

- capability of a product to perform specified functions under specified conditions for a specified period of time without interruptions and failures

Faultlessness: capability of a product to perform specified functions without fault under normal operation

Availability: capability of a product to be operational and accessible when required for use

Fault tolerance: capability of a product to operate as intended despite the presence of hardware or software faults

Recoverability: capability of a product in the event of an interruption or a failure to recover the data directly affected and re-establish the desired state of the system

FLEXIBILITY

- capability of a product to be adapted to changes in its requirements, contexts of use, or system environment

Adaptability: capability of a product to be effectively and efficiently adapted for or transferred to different hardware, software or other operational or usage environments

Scalability: capability of a product to handle growing or shrinking workloads or to adapt its capacity to handle variability

Installability: capability of a product to be effectively and efficiently installed successfully and/or uninstalled in a specified environment

Replaceability: capability of a product to replace another specified product for the same purpose in the same environment

The Research Monitoring System is designed to meet the ISO 25010 Reliability requirement by maintaining stable and consistent operation during normal use. The system includes basic input validation to reduce user errors and implements error-handling mechanisms that prevent the system from crashing when unexpected actions occur. For instance, when users submit forms or upload files that meet the required format, the system can act as a medium between advisor and their advisees. Errors aroused can be fix when occurred. These features help ensure that the system remains usable and dependable for students, advisers, and administrators during routine academic activities.

The system meets ISO 25010's Flexibility by being easy to change and improve as needs change. Administrators can update research requirements, deadlines, and user roles without rebuilding the whole system. Its simple and organized design also makes it easier to add new features in the future. This allows the Research Monitoring System to adjust over time and remain useful as research processes change.

METHODS AND TESTING PROCEDURE:

RELIABILITY & FLEXIBILITY

RELIABILITY

Unit Testing

Integration Testing

Error Handling Test

Stress and Load Testing

BackUp and Recovery Testing

Regression Testing

FLEXIBILITY

Configuration Testing

Modularity Testing

Scalability Testing

Scenario-based Testing

Documentation Review